



F2P Funnel Data — Candidate Data Dictionary

Read this before starting the case study. The CSV and view SQL will make more sense with this context.

Company context

PES is Office Timeline's subscription and billing management system — it's the source of record for subscriptions, seat assignments, and purchase history. References to `pes_subscriptions`, `pes_billingdata`, and related tables in the schemas all originate from this system.

BI tooling: Power BI is used for dashboards and business reporting. Databricks is the underlying data platform (SQL and notebooks). The data lake follows a medallion architecture: bronze (raw) → silver (cleaned, source-shaped) → gold (conformed facts and dimensions) → reporting (dashboard-specific views). The CSV you've been given was exported from the reporting layer.

What you're working with

The CSV is a full export of `reporting.view_funnel_dashboard`, Office Timeline's free-to-paid funnel tracking view. It contains one row per HubSpot contact — a person who interacted with OT's marketing funnel at some point between August 2025 and April 2026. The export contains roughly 104,000 rows.

One row = one HubSpot contact. A single company can have many contacts in the view. Whether that matters for your analysis is part of the exercise.

OT internal emails and known test addresses are already filtered out by the view. Everything else is included.

Field reference

Contact identity

Field	Type	Description
<code>email</code>	string	Anonymized contact email. Both the local part and the corporate domain are anonymized (e.g., <code>user-7a3f@company-abc12.com</code>). Personal email domains (gmail, yahoo, hotmail, etc.) are preserved as-is so personal vs. business contacts can be distinguished.
<code>first_contact_reference_date</code>	date	The date this contact was created in HubSpot. This is the funnel entry point — the date used for period filtering and conversion window calculations.

Contact source

These two fields describe how and why the contact entered HubSpot. They are the most important fields for understanding who is in the population.

Field	Type	Description
<code>contact_source_form</code>	string	The form or mechanism that created the contact. Two values represent people who chose to try the product themselves: <code>[Marketing] Freemium DL Sign-up</code> and <code>[Marketing] 14-day trial Sign-up</code> . All other values represent contacts created through other mechanisms.
<code>contact_source_type</code>	string	The channel category. <code>Integration</code> means the contact was created automatically by the billing system when a seat was assigned to that person — the individual did not visit a marketing page or fill out a form. Other values represent contacts created through standard marketing channels.

Field	Type	Description
<code>contact_source_traffic</code>	string	Traffic source at time of contact creation: <code>Paid Search</code> , <code>Organic Search</code> , <code>Direct</code> , <code>Other</code> . Based on HubSpot first-touch attribution (the channel of the contact's first-ever site visit), not last-click.

Note on `contact_source_type = 'Integration'`: When a company purchases a subscription and assigns seats to employees, OT's billing system automatically creates a HubSpot contact for each assigned user. These contacts did not independently choose to try the product. A product change (Account-Based Licensing, released November 17, 2025) was intended to reduce this — whether it did is something you can test.

Reseller flag

Field	Type	Description
<code>discriminator</code>	string	Identifies whether the purchase associated with this contact was made directly by the customer or through a reseller. Value <code>ResellerBillingInformation</code> means a reseller purchased on behalf of an end customer. Null or other values mean a direct purchase. Reseller contacts are present in the view.

Note on resellers: When a reseller purchases on behalf of a customer, the HubSpot contact may represent the reseller's employee or procurement team rather than an actual end user of the product.

Funnel progression flags

These flags track where each contact reached in the funnel. All are integers (0 or 1).

Field	Type	Description
<code>has_start_install</code>	int	1 if the contact started the product installation process.
<code>has_completed_install</code>	int	1 if the contact completed installation and had a working copy of the product.

Field	Type	Description
<code>has_purchased</code>	int	1 if a paid subscription purchase is attributed to this contact.
<code>has_license_created</code>	int	1 if a license (an alternative purchase pathway) was created for this contact.
<code>has_subscription_created</code>	int	1 if a subscription was created for this contact via a third pathway.
<code>is_buyer</code>	int	Label-based heuristic. Set to 1 if the contact's HubSpot labels contain <code>'resellers customer'</code> or <code>'owner'</code> . Computed independently of the three paid flags above. Whether this matches what you want as a paid indicator is part of the exercise.

The three paid flags (`has_purchased`, `has_license_created`, `has_subscription_created`) exist because OT has multiple purchase pathways that write to different backend systems. The dashboard's methodology note states `Paid = Completed a transaction after the contact creation date in HubSpot` — which maps to `has_purchased`. Whether that's the right indicator, or whether the analysis would benefit from a broader definition that includes the other pathways, is part of the exercise.

Conversion timing

Field	Type	Description
<code>first_purchase_date</code>	date	Date of first purchase, if <code>has_purchased = 1</code> . Null otherwise.
<code>license_created_at</code>	date	Date of license creation, if <code>has_license_created = 1</code> . Null otherwise.
<code>subscription_created_at</code>	date	Date of subscription creation, if <code>has_subscription_created = 1</code> . Null otherwise.

There is no pre-computed `days_to_paid` column. Compute conversion lag as:

```
DATEDIFF(COALESCE(first_purchase_date, license_created_at, subscription_created_at),
```

```
first_contact_reference_date)
```

Email domains

Emails in the CSV are anonymized. Personal email domains are preserved as-is; corporate domains are replaced with consistent hashes. Domains fall into two broad categories:

Personal domains — email providers used by individuals, not companies. These contacts cannot be attributed to a business. The following are the most common personal domains in the data:

gmail.com, yahoo.com, hotmail.com, outlook.com, live.com, icloud.com, aol.com, msn.com, protonmail.com, gmx.com, yandex.com, mail.com

Business domains — everything else. Corporate domains are anonymized in the CSV (e.g., company-abc12.com), but the anonymization is consistent: the same corporate domain in two different rows will hash to the same company-XXXXX.com string, so contacts can be grouped by domain. A single business domain may have many contacts across the funnel.

The current dashboard methodology

The current dashboard reports two conversion rates: **4.3% Contact:Paid** and **11.3% Complete:Paid**. The dashboard prominently highlights the 11.3% Complete:Paid number using the formula $\text{Conversion Rate} = \text{Paid} / \text{Complete}$. Both numbers are computed from this view with reseller transactions excluded; otherwise the methodology applied to compute them is left for you to investigate. The full SQL of the dashboard view is provided separately.

What's not in this dataset

- Subscription details (seat count, plan, price, renewal dates) — these live in a separate billing system
- Historical purchase records for companies that were already customers before entering the funnel

- Product usage data (activation, feature usage, session counts)
- Company-level attributes beyond what can be inferred from the email domain